

EOI Process Validation — Architecture & Data Model Documentation

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Table of Contents

- 1. Executive Summary
- 2. Architecture Overview
- 3. System Components
- 4. Entity Relationship Model
- 5. Field Descriptions
- 6. Process Flow
- 7. Validation Engine Design
- 8. Integration Patterns

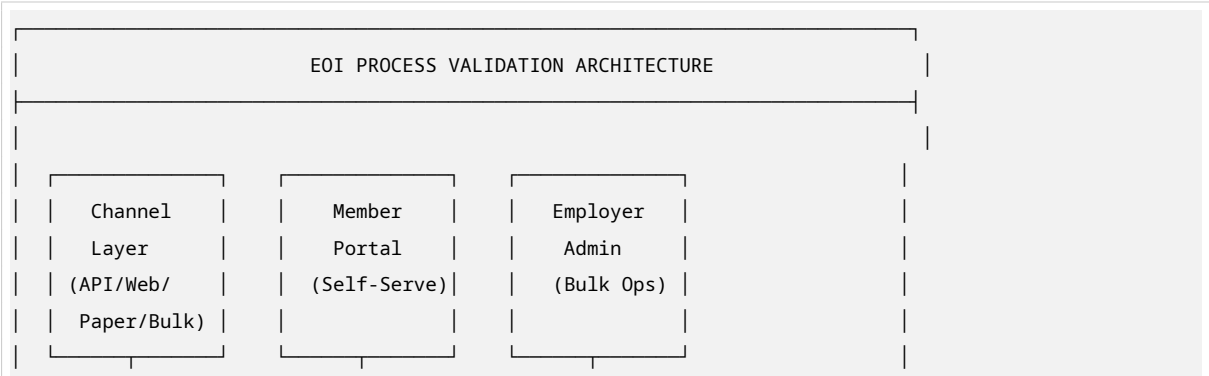
1. Executive Summary

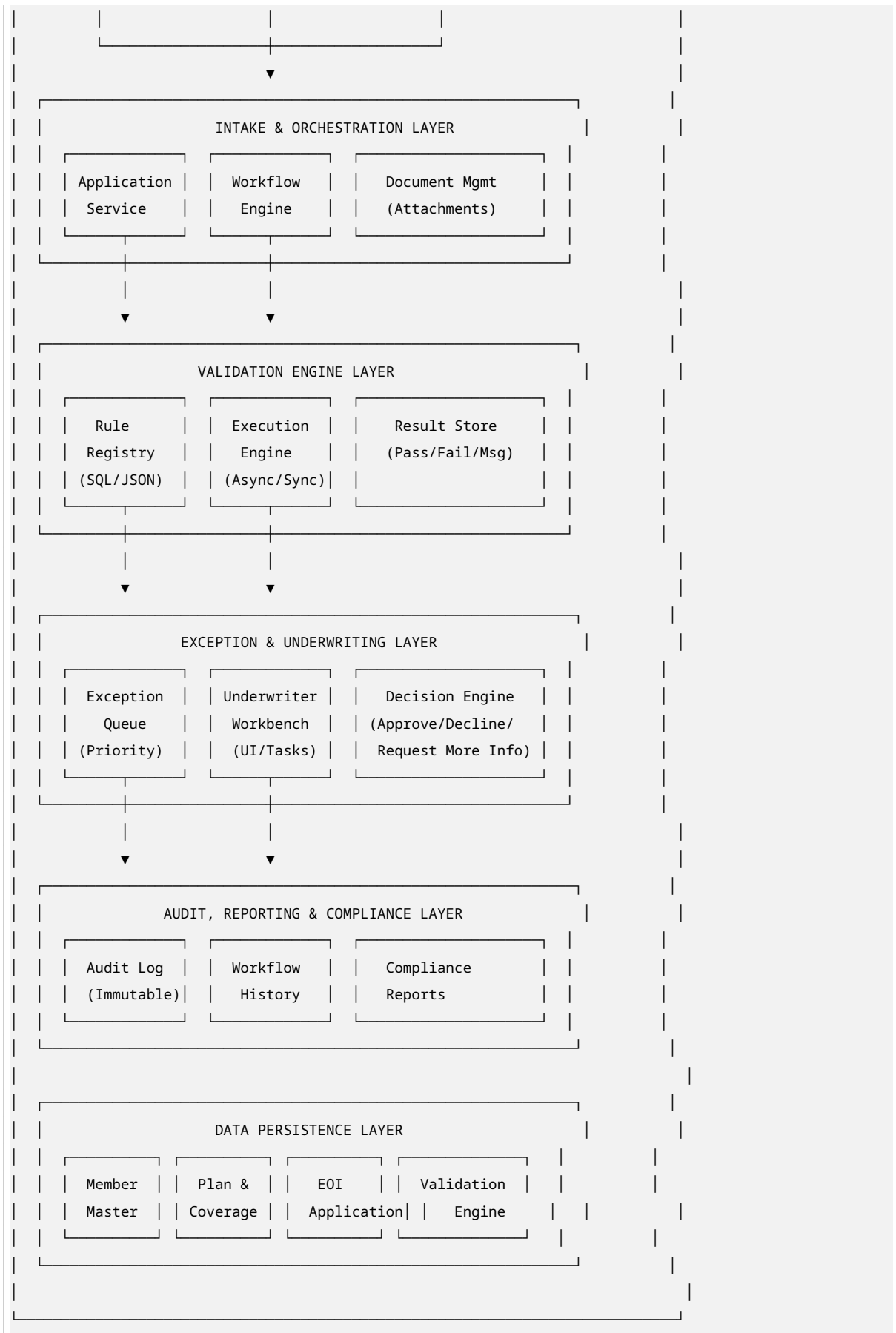
The **EOI (Evidence of Insurability) Process Validation System** is a rule-based automated underwriting platform designed to evaluate insurance coverage applications that exceed guaranteed issue limits. The system manages the complete lifecycle from application intake through validation, exception handling, underwriting decision, and audit compliance.

Key Capabilities

- **Automated Rule Validation:** Configurable validation rules with priority and severity levels
- **Exception Management:** Intelligent routing of failed validations to underwriter queues
- **Workflow Orchestration:** State-machine driven application lifecycle tracking
- **Audit & Compliance:** Comprehensive change logging for regulatory requirements
- **Hierarchical Data Model:** Employer → Division → Class → Member → Coverage chain

2. Architecture Overview





Architecture Principles

Principle	Description
Separation of Concerns	Validation logic is decoupled from workflow and data persistence
Configurability	Rules are data-driven, not hard-coded, enabling business user control
Auditability	Every state change and data mutation is logged immutably
Scalability	Async validation execution supports high-volume batch processing
Extensibility	Plugin architecture allows new rule categories without schema changes

3. System Components

3.1 Employer & Group Hierarchy

Manages the organizational structure: **Employer Group** → **Division** → **Class** → **Member**. This hierarchy determines eligibility, plan availability, and coverage rules.

3.2 Member Management

Central registry of insured members with demographic data, employment status, and linkage to group hierarchy. Tracks eligibility separately from enrollment.

3.3 Insurance Plan & Coverage

Defines available insurance products (Life, Disability, Critical Illness) with coverage parameters. **Member_Coverage** links members to active plans with specific amounts.

3.4 EOI Application

The core transaction entity representing a request for coverage above guaranteed issue limits. Tracks status through the entire lifecycle.

3.5 Validation Engine

Rule Registry: Stores configurable business rules with expressions, priority, and severity.

Execution Engine: Runs rules against application data and produces pass/fail results.

Result Store: Persists detailed validation outcomes for reporting and exception routing.

3.6 Exception & Underwriting

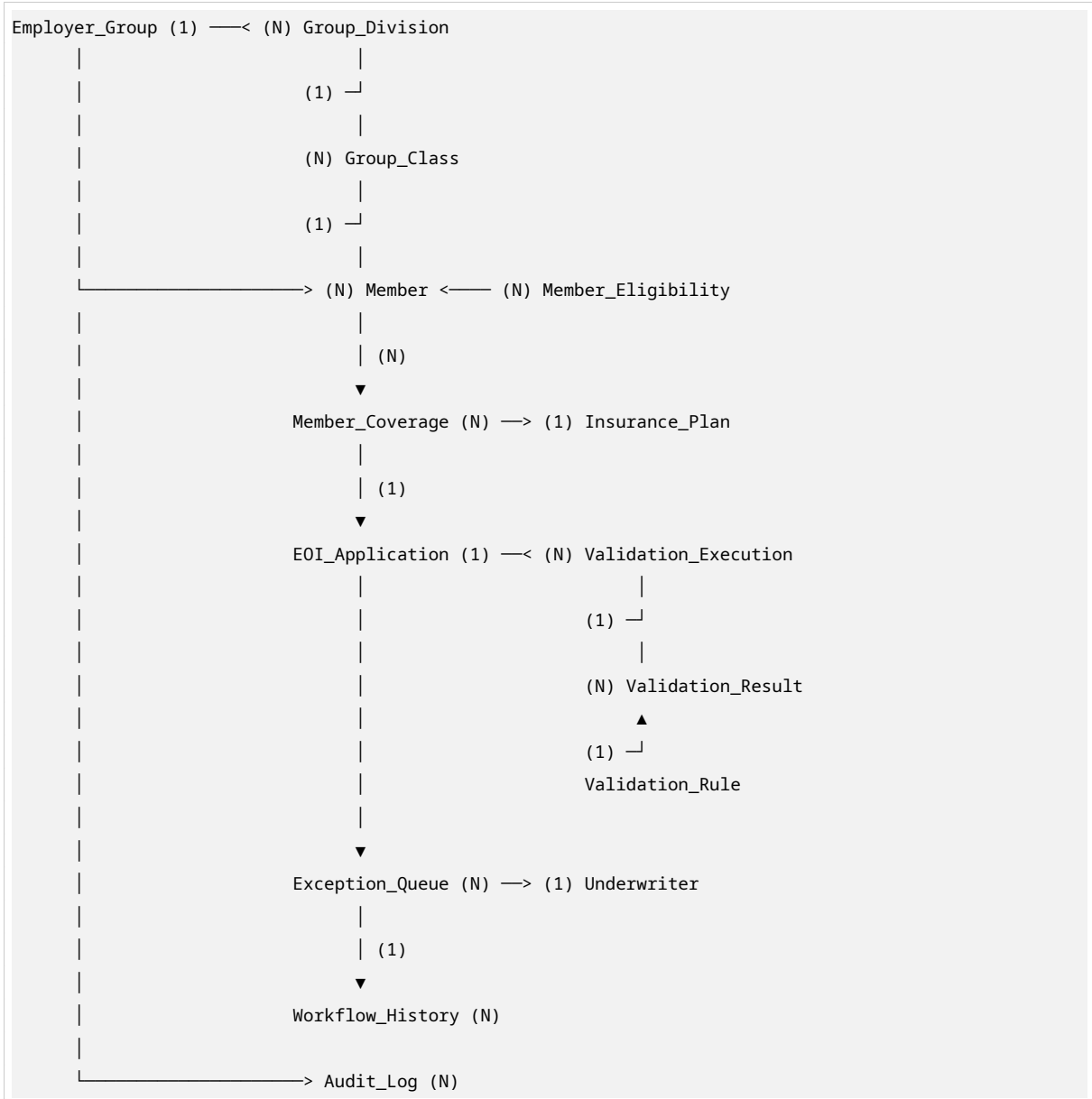
Failed validations create exceptions in a priority queue. Underwriters claim, review, and resolve exceptions through a dedicated workbench.

3.7 Workflow & Audit

State machine tracks application status transitions. Immutable audit logs capture all data changes for compliance (HIPAA, SOX, state insurance regulations).

4. Entity Relationship Model

4.1 Relationship Diagram (Textual)



4.2 Cardinality Summary

Parent Entity	Child Entity	Cardinality	Relationship
Employer_Group	Group_Division	1:N	One group has many divisions

Table 2 – continued

Parent Entity	Child Entity	Cardinality	Relationship
Group_Division	Group_Class	1:N	One division has many classes
Employer_Group	Member	1:N	One group has many members
Group_Division	Member	1:N	One division has many members (optional)
Group_Class	Member	1:N	One class has many members (optional)
Member	Member_Eligibility	1:N	One member has eligibility history
Member	Member_Coverage	1:N	One member has multiple coverages
Insurance_Plan	Member_Coverage	1:N	One plan covers many members
Member	EOI_Application	1:N	One member submits many applications
Insurance_Plan	EOI_Application	1:N	One plan has many applications
EOI_Application	Validation_Execution	1:N	One app has many validation runs
Validation_Execution	Validation_Result	1:N	One execution produces many results
Validation_Rule	Validation_Result	1:N	One rule appears in many results
EOI_Application	Exception_Queue	1:N	One app may have multiple exceptions
Underwriter	Exception_Queue	1:N	One underwriter handles many exceptions
EOI_Application	Workflow_History	1:N	One app has many status transitions
EOI_Application	Audit_Log	1:N	One app generates many audit entries

5. Field Descriptions

5.1 Employer_Group

Field	Type	Nullable	Description
group_id	BIGINT	NO	Primary key. Unique identifier for the employer group.

Table 3 – continued

Field	Type	Nullable	Description
group_number	VARCHAR(50)	NO	External reference number assigned by the carrier or administrator. Must be unique.
group_name	VARCHAR(255)	NO	Legal or trading name of the employer organization.
employer_name	VARCHAR(255)	NO	Name of the employing entity (may differ from group_name in trust arrangements).
group_type	VARCHAR(50)	NO	Classification: Corporate , Small Business , Association , Union , Government .
market_segment	VARCHAR(50)	YES	Market categorization: Commercial , Individual , Government , Voluntary .
status	VARCHAR(20)	NO	Lifecycle state: ACTIVE , INACTIVE , PENDING , TERMINATED .
effective_date	DATE	NO	Date when the group contract becomes effective.
termination_date	DATE	YES	Date when the group contract ends. NULL if ongoing.
created_date	TIMESTAMP	NO	Row creation timestamp. Auto-populated.
updated_date	TIMESTAMP	NO	Last modification timestamp. Auto-updated.

5.2 Group_Division

Field	Type	Nullable	Description
division_id	BIGINT	NO	Primary key. Unique identifier for the division.
group_id	BIGINT	NO	Foreign key to Employer_Group. Establishes ownership.
division_code	VARCHAR(50)	NO	Short code for the division (e.g., DIV-001 , EAST_REGION).
division_name	VARCHAR(255)	NO	Descriptive name (e.g., “Northeast Sales Division”).
effective_date	DATE	NO	When this division becomes active under the group contract.
status	VARCHAR(20)	NO	ACTIVE , INACTIVE , or TERMINATED .

5.3 Group_Class

Field	Type	Nullable	Description
class_id	BIGINT	NO	Primary key. Unique identifier for the class.
division_id	BIGINT	NO	Foreign key to Group_Division. Classes belong to divisions.
class_code	VARCHAR(50)	NO	Short code (e.g., CLS-A, EXECUTIVE).
class_name	VARCHAR(255)	NO	Descriptive name (e.g., “Executive Officers”, “Hourly Workers”).
eligibility_rule	TEXT	YES	JSON or expression defining who qualifies (e.g., minimum hours, job grade).
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, or TERMINATED.

5.4 Member

Field	Type	Nullable	Description
member_id	BIGINT	NO	Primary key. System-generated unique member identifier.
employee_number	VARCHAR(50)	NO	Employer’ s internal employee ID or badge number.
first_name	VARCHAR(100)	NO	Member’ s legal first name.
last_name	VARCHAR(100)	NO	Member’ s legal last name.
dob	DATE	NO	Date of Birth. Used for age-based eligibility and rate calculations.
gender	CHAR(1)	YES	M=Male, F=Female, O=Other, U=Unknown.
ssn	VARCHAR(11)	YES	Social Security Number. Must be encrypted at rest (AES-256).
email	VARCHAR(255)	YES	Primary email for notifications and portal access.
phone	VARCHAR(20)	YES	Primary contact phone number.
employment_status	VARCHAR(30)	NO	ACTIVE, TERMINATED, LEAVE, RETIRED. Drives eligibility.
hire_date	DATE	NO	Original date of hire. Used for waiting period calculations.
group_id	BIGINT	NO	Foreign key to Employer_Group. Required association.
division_id	BIGINT	YES	Foreign key to Group_Division. Optional sub-organization.
class_id	BIGINT	YES	Foreign key to Group_Class. Optional benefit class.
status	VARCHAR(20)	NO	Member record status: ACTIVE, INACTIVE, TERMINATED.

5.5 Member_Eligibility

Field	Type	Nullable	Description
eligibility_id	BIGINT	NO	Primary key. Unique eligibility determination record.
member_id	BIGINT	NO	Foreign key to Member. Links to the subject.
effective_date	DATE	NO	Start date of this eligibility period.
termination_date	DATE	YES	End date of eligibility. NULL if currently eligible.
eligibility_status	VARCHAR(30)	NO	ELIGIBLE, INELIGIBLE, PENDING, EXPIRED.
reason	TEXT	YES	Human-readable explanation for the determination.
verified_date	TIMESTAMP	YES	When eligibility was last verified by a system or user.
verified_by	VARCHAR(100)	YES	User ID or system process that performed verification.
created_date	TIMESTAMP	NO	Record creation timestamp.
updated_date	TIMESTAMP	NO	Last update timestamp.

5.6 Insurance_Plan

Field	Type	Nullable	Description
plan_id	BIGINT	NO	Primary key. Unique plan identifier.
plan_code	VARCHAR(50)	NO	Carrier-specific plan code (e.g., GL-2026-BASIC). Unique.
plan_name	VARCHAR(255)	NO	Marketing or descriptive name (e.g., “Group Life Basic”).
product_type	VARCHAR(50)	NO	Insurance line: Life, Disability, Critical Illness, Accident.
coverage_type	VARCHAR(50)	NO	Tier: Basic, Supplemental, Voluntary, Core.
guaranteed_issue	BOOLEAN	YES	If TRUE, no EOI required up to the plan’s GI limit.
maximum_coverage	DECIMAL(15,2)	YES	Absolute maximum coverage amount available under this plan.
minimum_coverage	DECIMAL(15,2)	YES	Minimum required enrollment amount. Default 0.
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, TERMINATED.

5.7 Member_Coverage

Field	Type	Nullable	Description
coverage_id	BIGINT	NO	Primary key. Unique coverage enrollment record.
member_id	BIGINT	NO	Foreign key to Member. The insured individual.
plan_id	BIGINT	NO	Foreign key to Insurance_Plan. The enrolled product.
coverage_amount	DECIMAL(15,2)	NO	Dollar amount of coverage (e.g., \$150,000.00).
coverage_level	VARCHAR(50)	YES	Tier: Employee, Employee+Spouse, Employee+Child, Family.
coverage_status	VARCHAR(20)	NO	ACTIVE, PENDING, TERMINATED, SUSPENDED.
effective_date	DATE	NO	When coverage becomes active.
termination_date	DATE	YES	When coverage ends. NULL if active.

5.8 EOI_Application

Field	Type	Nullable	Description
application_id	BIGINT	NO	Primary key. System-generated application ID.
application_number	VARCHAR(50)	NO	Business-facing reference number (e.g., EOI-2026-0004521). Unique.
member_id	BIGINT	NO	Foreign key to Member. Applicant.
plan_id	BIGINT	NO	Foreign key to Insurance_Plan. Target plan.
requested_amount	DECIMAL(15,2)	NO	Additional coverage amount requested above current/GI limit.
application_date	DATE	NO	Date the application was submitted.
submission_channel	VARCHAR(50)	NO	Origin: PORTAL, PAPER, EMAIL, API, BULK_IMPORT.
application_status	VARCHAR(30)	NO	Business outcome: SUBMITTED, UNDER_REVIEW, APPROVED, DECLINED, PENDING_INFO, WITHDRAWN.
workflow_status	VARCHAR(30)	NO	Process stage: INTAKE, VALIDATION, UNDERWRITING, DECISION, CLOSED, EXCEPTION.
created_by	VARCHAR(100)	YES	User ID or system that created the application.
created_date	TIMESTAMP	NO	Submission timestamp.

5.9 Validation_Rule

Field	Type	Nullable	Description
rule_id	BIGINT	NO	Primary key. Unique rule identifier.
rule_name	VARCHAR(255)	NO	Human-readable name (e.g., “Age Under 70 at Application”). Unique.
rule_category	VARCHAR(50)	NO	Functional area: ELIGIBILITY, COVERAGE_LIMIT, DATA_QUALITY, COMPLIANCE, BUSINESS_RULE.
rule_expression	TEXT	NO	The validation logic. Can be SQL, JSON Logic, or DSL reference.
priority	INT	NO	Execution order. Lower numbers run first (1 = highest).
severity	VARCHAR(20)	NO	Impact level: INFO, WARNING, ERROR, CRITICAL.
enabled	BOOLEAN	YES	If FALSE, rule is skipped during execution.
created_date	TIMESTAMP	NO	When the rule was defined.

5.10 Validation_Execution

Field	Type	Nullable	Description
execution_id	BIGINT	NO	Primary key. Unique validation run identifier.
application_id	BIGINT	NO	Foreign key to EOI_Application. The subject application.
started_time	TIMESTAMP	NO	When the validation run began.
completed_time	TIMESTAMP	YES	When the validation run finished. NULL if still running.
execution_status	VARCHAR(30)	NO	RUNNING, COMPLETED, FAILED, TIMEOUT.
rule_engine_version	VARCHAR(20)	YES	Semantic version of the validation engine (e.g., 2.4.1).

5.11 Validation_Result

Field	Type	Nullable	Description
result_id	BIGINT	NO	Primary key. Unique result record.
execution_id	BIGINT	NO	Foreign key to Validation_Execution. Parent run.
rule_id	BIGINT	NO	Foreign key to Validation_Rule. The rule evaluated.

Table 13 – continued

Field	Type	Nullable	Description
passed	BOOLEAN	NO	TRUE if the application satisfied the rule; FALSE if it failed.
expected_value	TEXT	YES	The value or condition the rule expected (for debugging).
actual_value	TEXT	YES	The actual value found in the application data.
message	TEXT	YES	Human-readable explanation of the result (success or failure).
created_date	TIMESTAMP	NO	When the result was recorded.

5.12 Underwriter

Field	Type	Nullable	Description
underwriter_id	BIGINT	NO	Primary key. Unique underwriter identifier.
employee_id	VARCHAR(50)	NO	HR system employee ID. Unique.
name	VARCHAR(255)	NO	Full name of the underwriter.
email	VARCHAR(255)	NO	Business email for notifications and SSO.
role	VARCHAR(50)	NO	Seniority: JUNIOR, SENIOR, LEAD, MANAGER. Determines authority limits.
department	VARCHAR(100)	YES	Organizational unit (e.g., “Group Life Underwriting”).
status	VARCHAR(20)	NO	ACTIVE, INACTIVE, ON_LEAVE.

5.13 Exception_Queue

Field	Type	Nullable	Description
exception_id	BIGINT	NO	Primary key. Unique exception record.
application_id	BIGINT	NO	Foreign key to EOI_Application. The problematic application.
exception_type	VARCHAR(50)	NO	Category: VALIDATION_FAILURE, MANUAL_REVIEW, SYSTEM_ERROR, POLICY_EXCEPTION.
priority	INT	NO	Triage level: 1=Critical, 2=High, 3=Medium, 4=Low.
assigned_underwriter_id	BIGINT	YES	Foreign key to Underwriter. NULL if unassigned.

Table 15 – continued

Field	Type	Nullable	Description
status	VARCHAR(20)	NO	OPEN, IN_PROGRESS, RESOLVED, ESCALATED.
created__date	TIMESTAMP	NO	When the exception was generated.
resolved__date	TIMESTAMP	YES	When the exception was closed. NULL until resolved.

5.14 Workflow_History

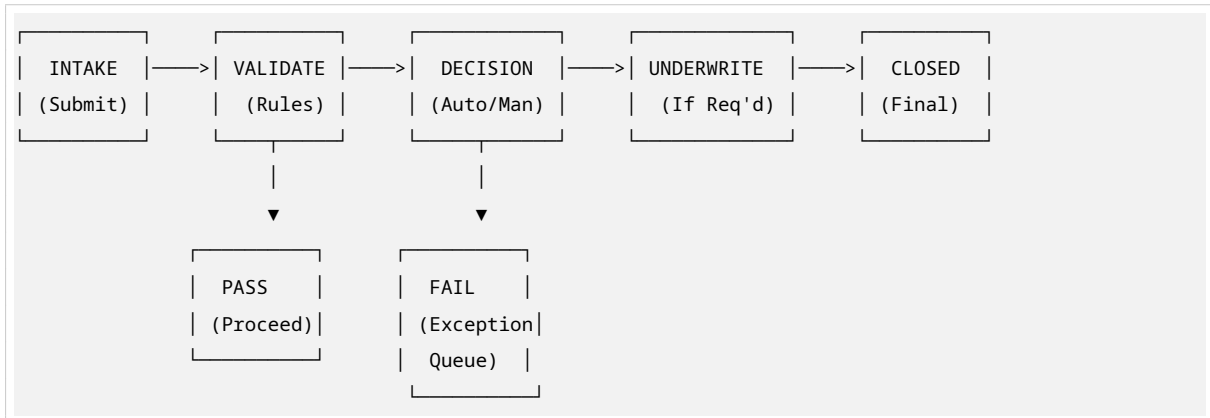
Field	Type	Nullable	Description
workflow__id	BIGINT	NO	Primary key. Unique transition record.
application__id	BIGINT	NO	Foreign key to EOI_Application. The subject.
previous__status	VARCHAR(30)	NO	Status before the transition.
current__status	VARCHAR(30)	NO	Status after the transition.
changed__by	VARCHAR(100)	NO	User ID or SYSTEM that triggered the change.
changed__date	TIMESTAMP	NO	Exact timestamp of the transition.
comments	TEXT	YES	Free-text explanation for the status change.

5.15 Audit_Log

Field	Type	Nullable	Description
audit__id	BIGINT	NO	Primary key. Immutable audit record identifier.
application__id	BIGINT	YES	Foreign key to EOI_Application. Optional if change is unrelated to a specific app.
table__name	VARCHAR(100)	NO	Database table that was modified.
field__name	VARCHAR(100)	NO	Specific column that changed.
old__value	TEXT	YES	Previous value (NULL for inserts).
new__value	TEXT	YES	New value (NULL for deletes).
changed__by	VARCHAR(100)	NO	Identity of the actor (user or system process).
changed__date	TIMESTAMP	NO	Timestamp of the change. Immutable.

6. Process Flow

6.1 Standard EOI Application Lifecycle



6.2 Detailed Process Steps

Step	Stage	Actor	Action	Outcome
1	Intake	Member/Admin	Submit EOI application with requested amount	Record created; workflow_status = INTAKE
2	Pre-Validation	System	Check member eligibility and plan availability	If invalid, app declined immediately
3	Rule Execution	Validation Engine	Execute all enabled rules by priority	Validation_Execution + Validation_Result records created
4	Result Analysis	System	Aggregate results; check for ERROR/CRITICAL failures	If all pass → auto-approve or route to underwriting based on amount
5	Exception Routing	System	Create Exception_Queue entry for failed rules	Priority assigned based on severity and amount
6	Underwriter Review	Underwriter	Claim exception, review application, request info if needed	Status updated to IN_PROGRESS
7	Decision	Underwriter/ System	Approve, decline, or pend for more information	application_status updated; Workflow_History record created
8	Coverage Update	System	If approved, update Member_Coverage with new amount	Member record synchronized
9	Closure	System	Archive application; send notification	workflow_status = CLOSED

7. Validation Engine Design

7.1 Rule Categories

Category	Purpose	Example Rules
ELIGIBILITY	Verify member qualifies for EOI	Member is ACTIVE; hire date > 90 days ago
COVERAGE_LIMIT	Ensure requested amount is within plan bounds	Requested <= Plan.maximum_coverage; Requested >= GI limit
DATA_QUALITY	Validate completeness and format	SSN is valid; Email is present; DOB is not future date
COMPLIANCE	Regulatory and policy checks	Age at application < 70; State allows product type
BUSINESS_RULE	Carrier-specific underwriting criteria	No prior declinations in 12 months; Coverage increase < 3x current

7.2 Severity Levels

Severity	Behavior	Example
INFO	Logged only; no impact on routing	“Member has multiple active coverages”
WARNING	Logged; may trigger review if combined with other issues	“Requested amount is 90% of maximum”
ERROR	Fails validation; routes to exception queue	“Member is not currently eligible”
CRITICAL	Immediate decline; no underwriter review	“Fraud indicator detected”

7.3 Execution Model

Rules are executed sequentially by **priority** (ascending). Execution stops early only for CRITICAL failures (hard stop). All other severities continue to collect full results for comprehensive reporting.

```
FOR each enabled_rule IN rules ORDER BY priority ASC:
    result = evaluate(rule.expression, application_context)
    save_result(result)
    IF rule.severity == 'CRITICAL' AND result.passed == FALSE:
        mark_immediate_decline()
        BREAK
```

8. Integration Patterns

8.1 Inbound Integrations

Source	Pattern	Data
Member Portal	REST API	Application submission, document upload
Employer HR System	SFTP/Batch	Member demographics, eligibility files
Underwriter Workbench	Web UI + API	Decisions, exception resolution
Document Management	Event-driven	Medical records, attestation forms

8.2 Outbound Integrations

Target	Pattern	Data
Member Portal	Webhook/Push	Status updates, decision notifications
Email Service	SMTP/API	Approval/decline letters, information requests
Policy Admin System	REST/SOAP	Approved coverage enrollments
Data Warehouse	ETL/Batch	Analytics, regulatory reporting
SIEM/Monitoring	Event Stream	Security events, anomaly detection

8.3 Event-Driven Architecture

Key domain events published to message bus:

- EOI.ApplicationSubmitted
- EOI.ValidationStarted
- EOI.ValidationCompleted
- EOI.ExceptionCreated
- EOI.DecisionMade
- EOI.CoverageActivated
- EOI.WorkflowStatusChanged

Appendix A: Data Retention & Compliance

Entity	Retention Period	Encryption	PII
Member	7 years post-termination	AES-256	Yes (SSN, DOB)
EOI_Application	10 years	AES-256	Yes
Audit_Log	10 years (immutable)	AES-256	Yes
Validation_Result	7 years	AES-256	No

Table 23 – continued

Entity	Retention Period	Encryption	PII
Exception_Queue	7 years	AES-256	Yes

Appendix B: Glossary

Term	Definition
EOI	Evidence of Insurability –medical or financial proof required for coverage above standard limits
Guaranteed Issue (GI)	Coverage amount available without health questions or underwriting
Underwriter	Insurance professional who evaluates risk and makes coverage decisions
Exception Queue	Worklist of applications requiring manual review due to validation failures
Rule Engine	Software component that evaluates business rules against application data
Workflow Status	Technical process stage (Intake -> Validation -> Underwriting -> Decision -> Closed)
Application Status	Business outcome state (Submitted, Approved, Declined, etc.)

Document generated from EOI_Validation_Process entity data model. For questions, contact the Architecture Team.